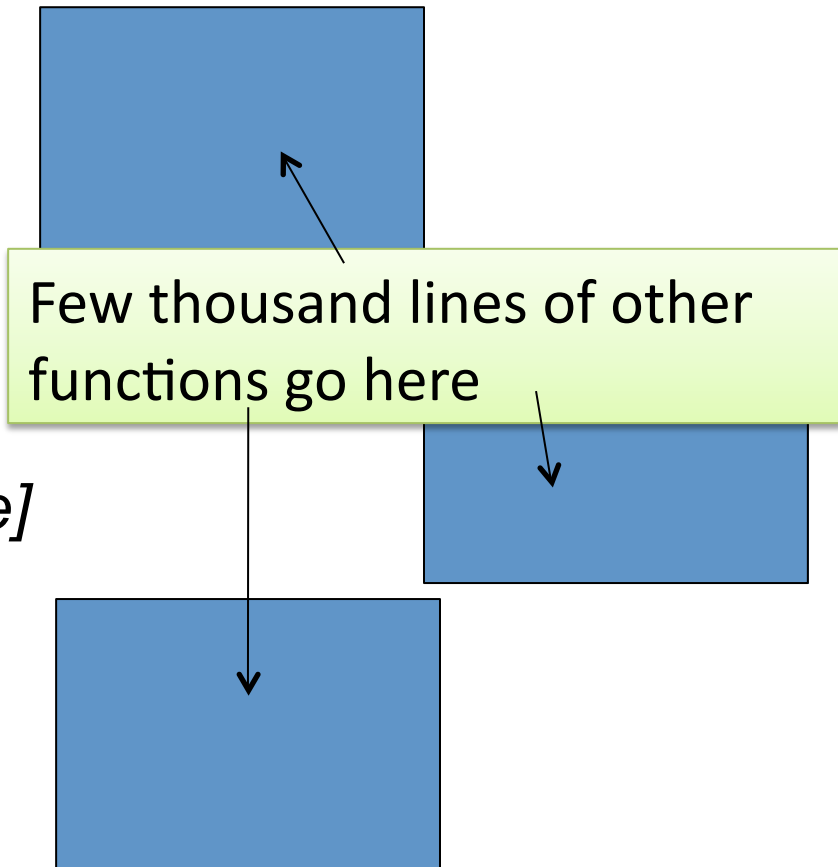


Basic fMRI, Basic Decoding, & TDT – The Decoding Toolbox

Kai Görgen
BCCN Berlin

IK2012, March 20th, 2012

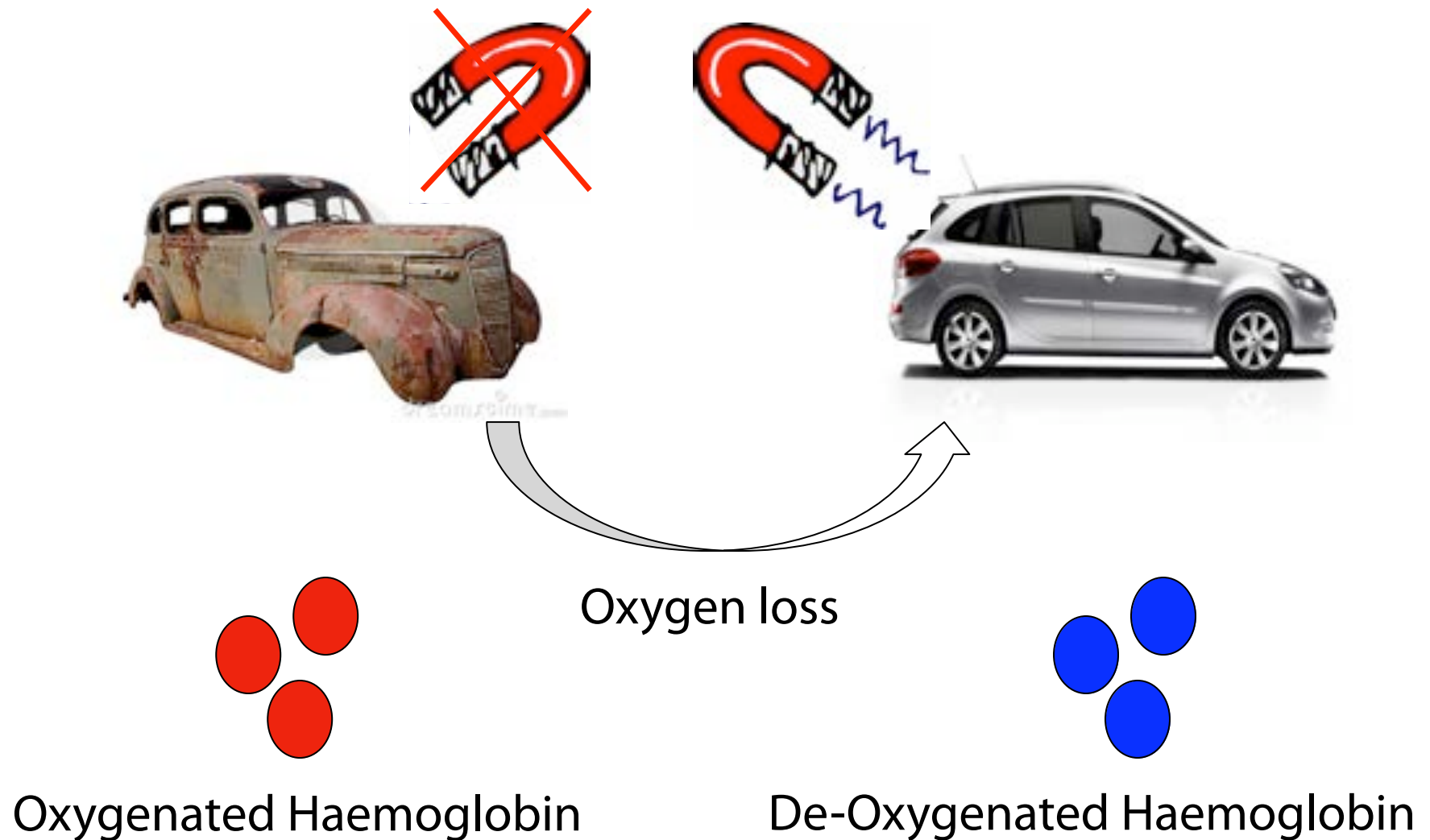
```
for runs = 1:nruns
    ...
    [78 lines of code go here]
    ...
    for set = 1:nsets
        ...
        [258 lines of code go here]
        ...
        output = outputs(1:set)
    end
    % output = outputs(1:set);
end
```



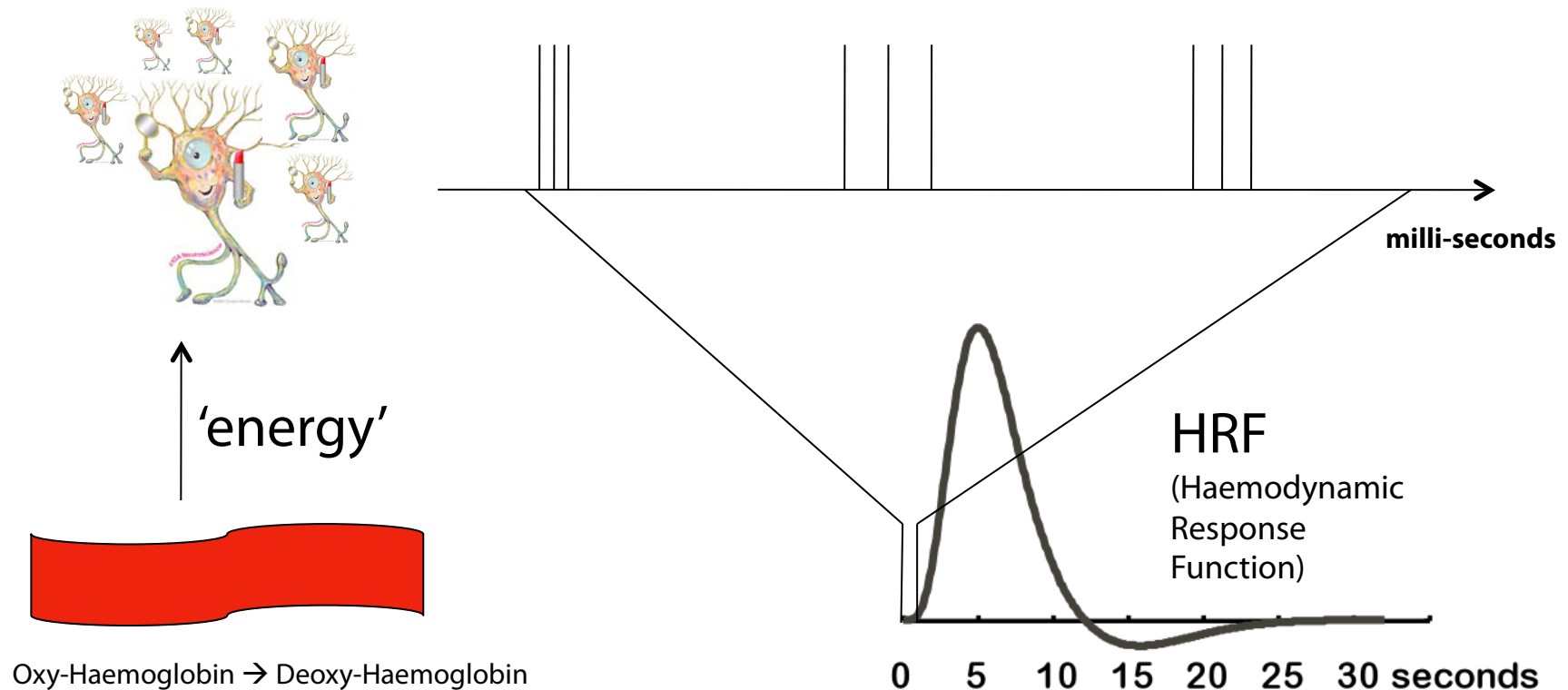
Outline

- Background
 - Very short: fMRI
 - Decoding
 - What and why
 - What to expect
 - A Very short overview about The Decoding Toolbox
- Hands-On

A word on fMRI

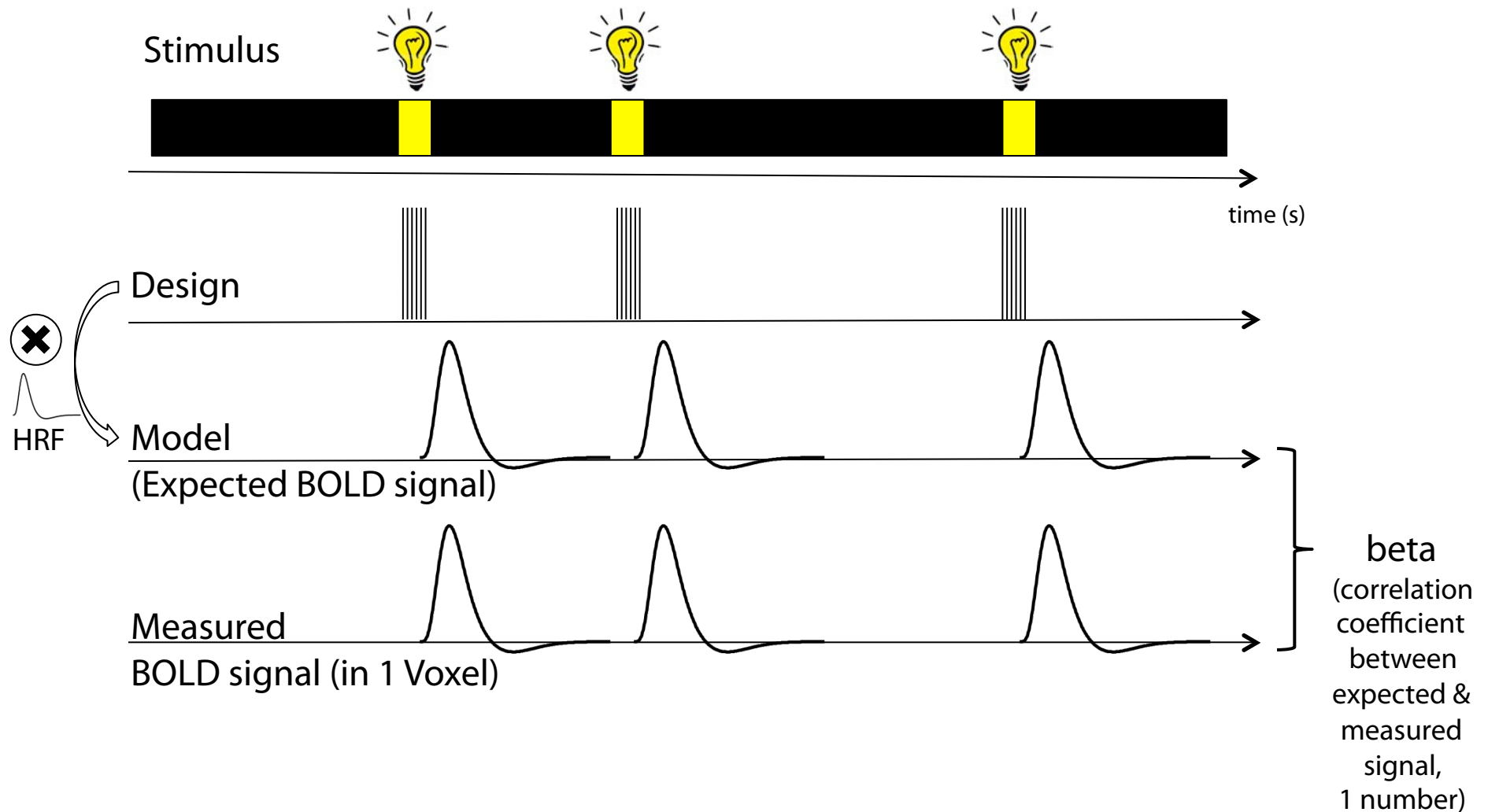


A word on fMRI

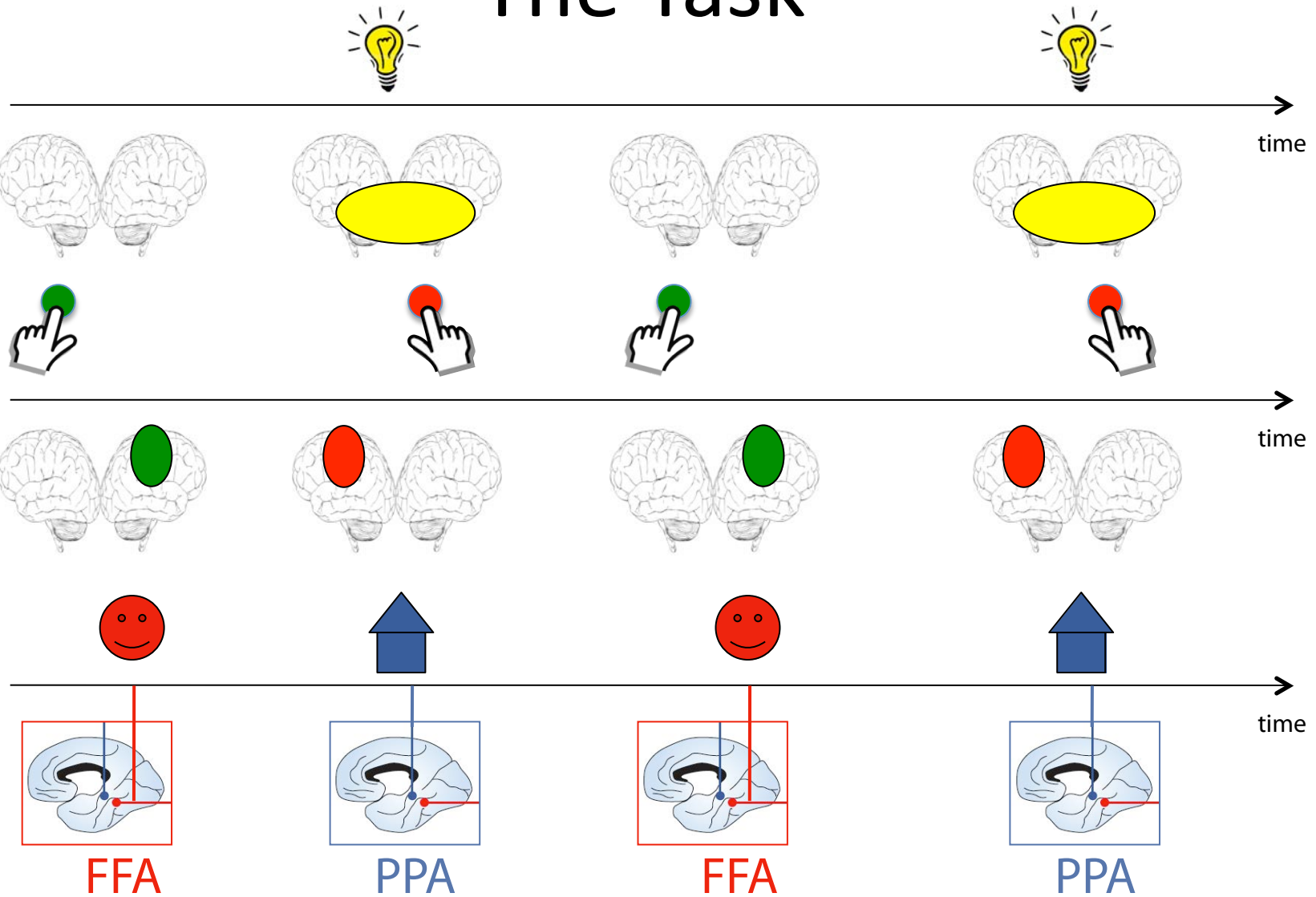


Oxygen unbound ('un'-rost) → Magnetic → 'BOLD' signal

Modelling BOLD & Betas



The Task



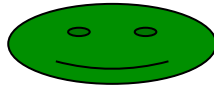
Kanwisher et al, 95, JNeuro

Eppstein et al, 98, Science

The Task



FFA!

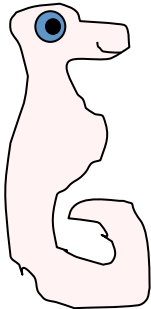


Big-Green-Face Area?

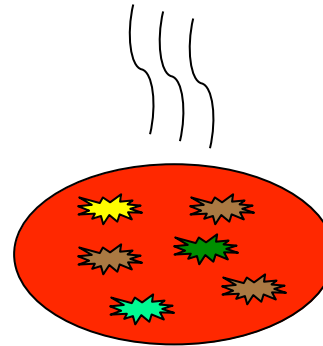
Blue-Slim-Face Area?



Uncle -Fred's-Face Area?



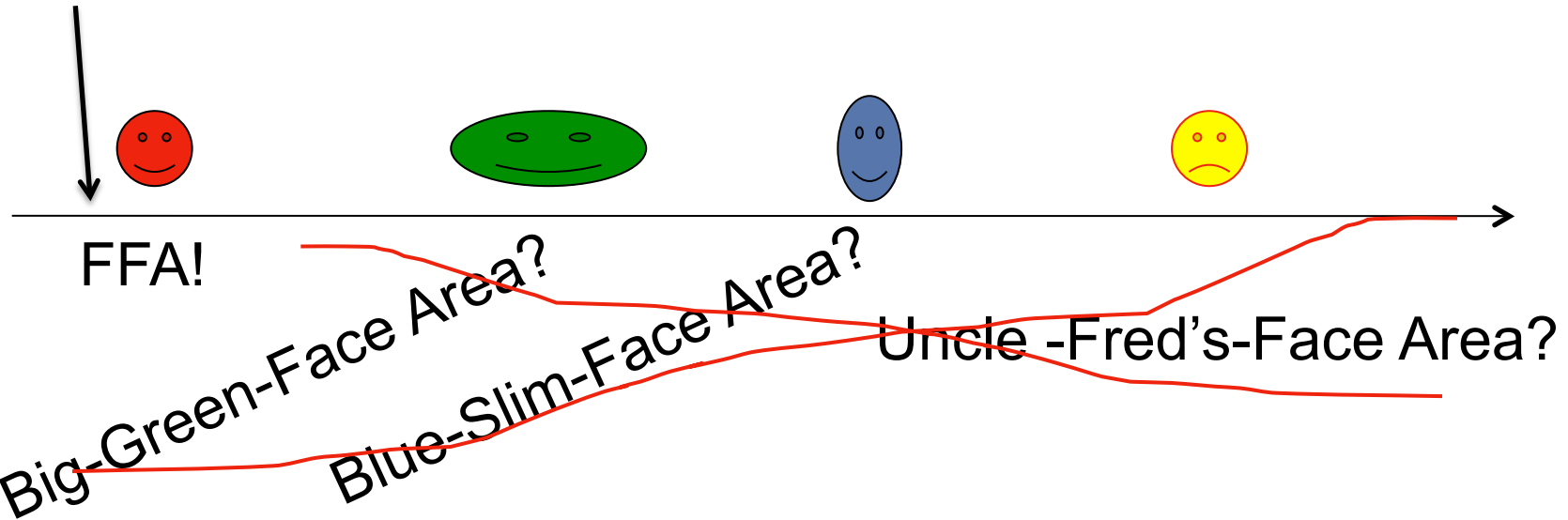
Badly-Drawn-Pink-Sea-Horse Area?



Günne-Pizza-Soup Area?

The Task

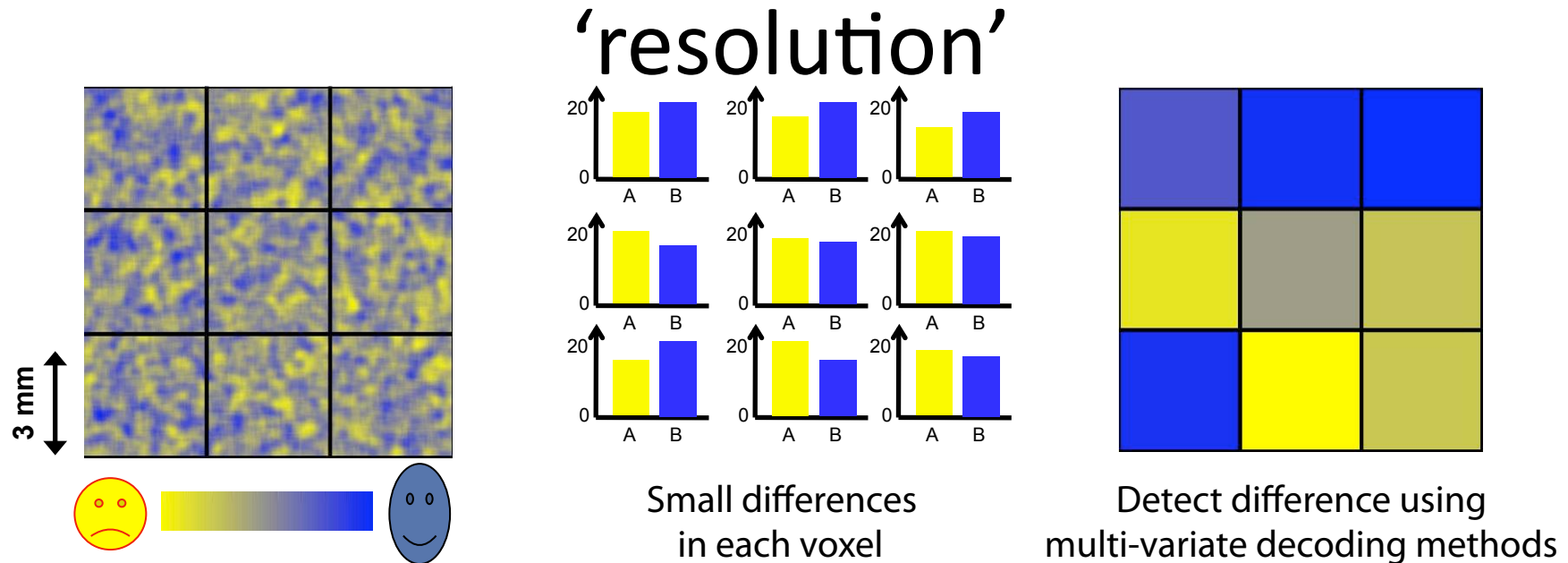
More?



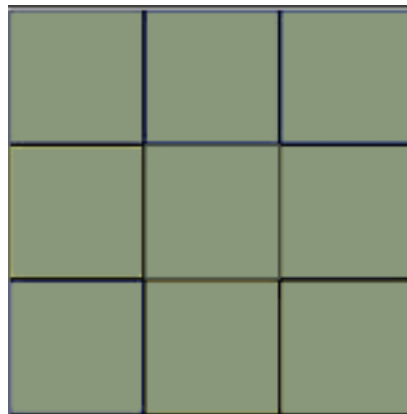
Idea: Different **patterns** of neural activity ***within*** areas encode the difference between different faces (objects/thoughts/...)!

Objective for decoding: Find these areas

How decoding improves fMRI



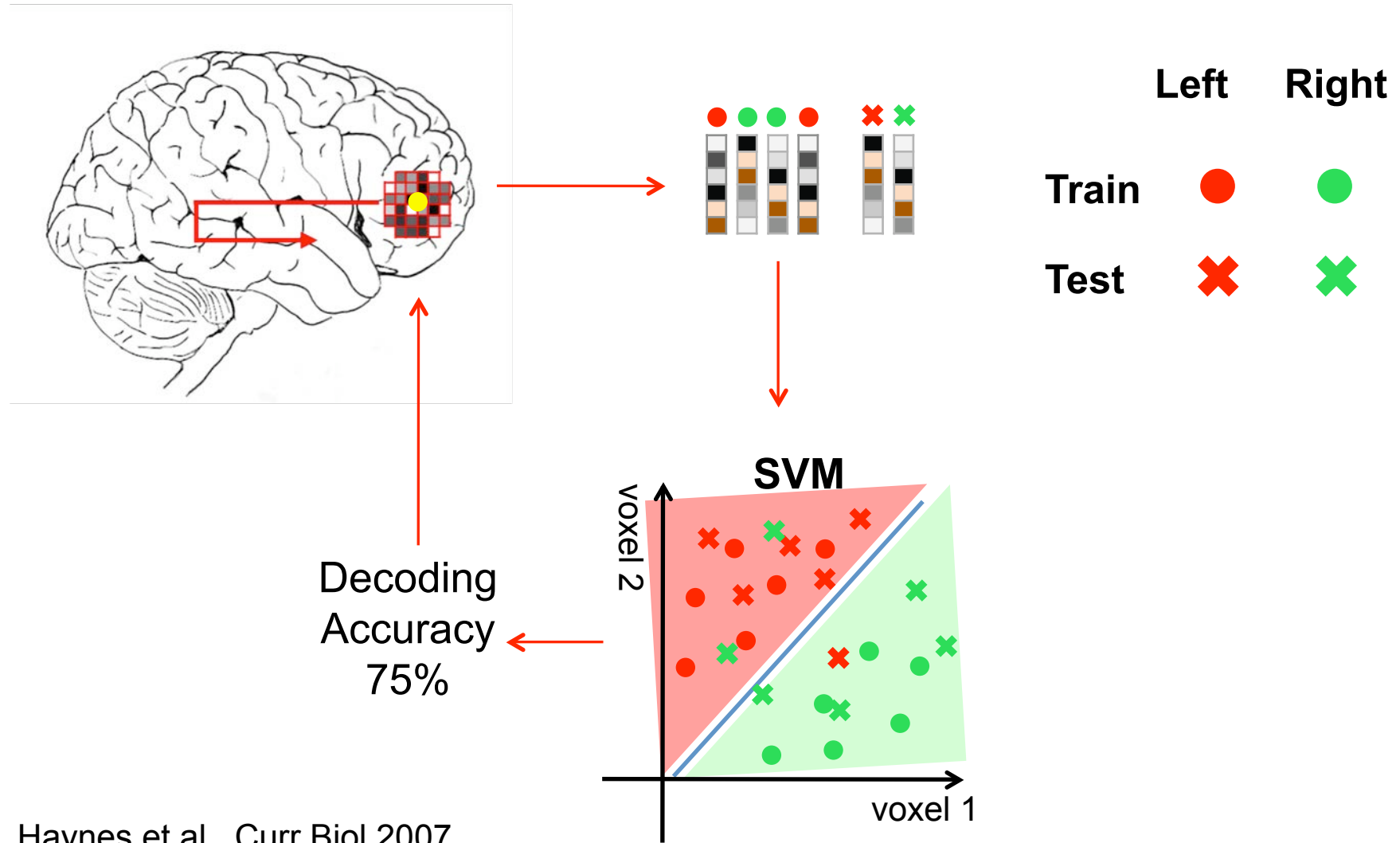
Conventional fMRI:
Smoothing to get rid of noise
→ No difference



Early decoding studies

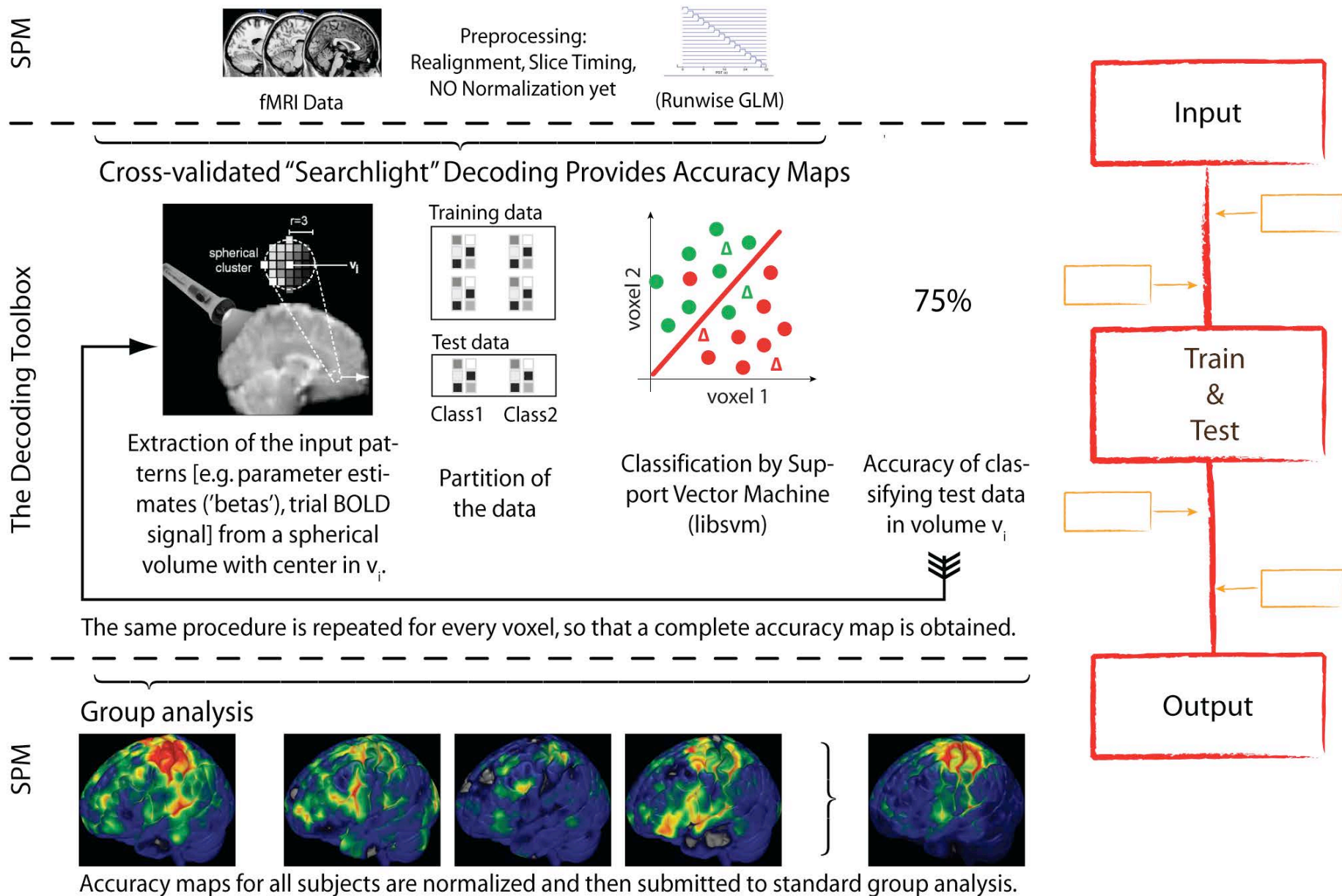
- Haxby
- Kamatami & Tong
- Haynes & Rees

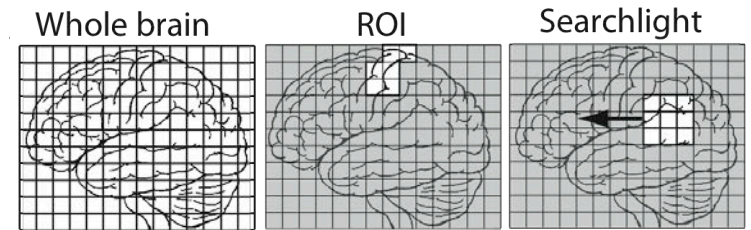
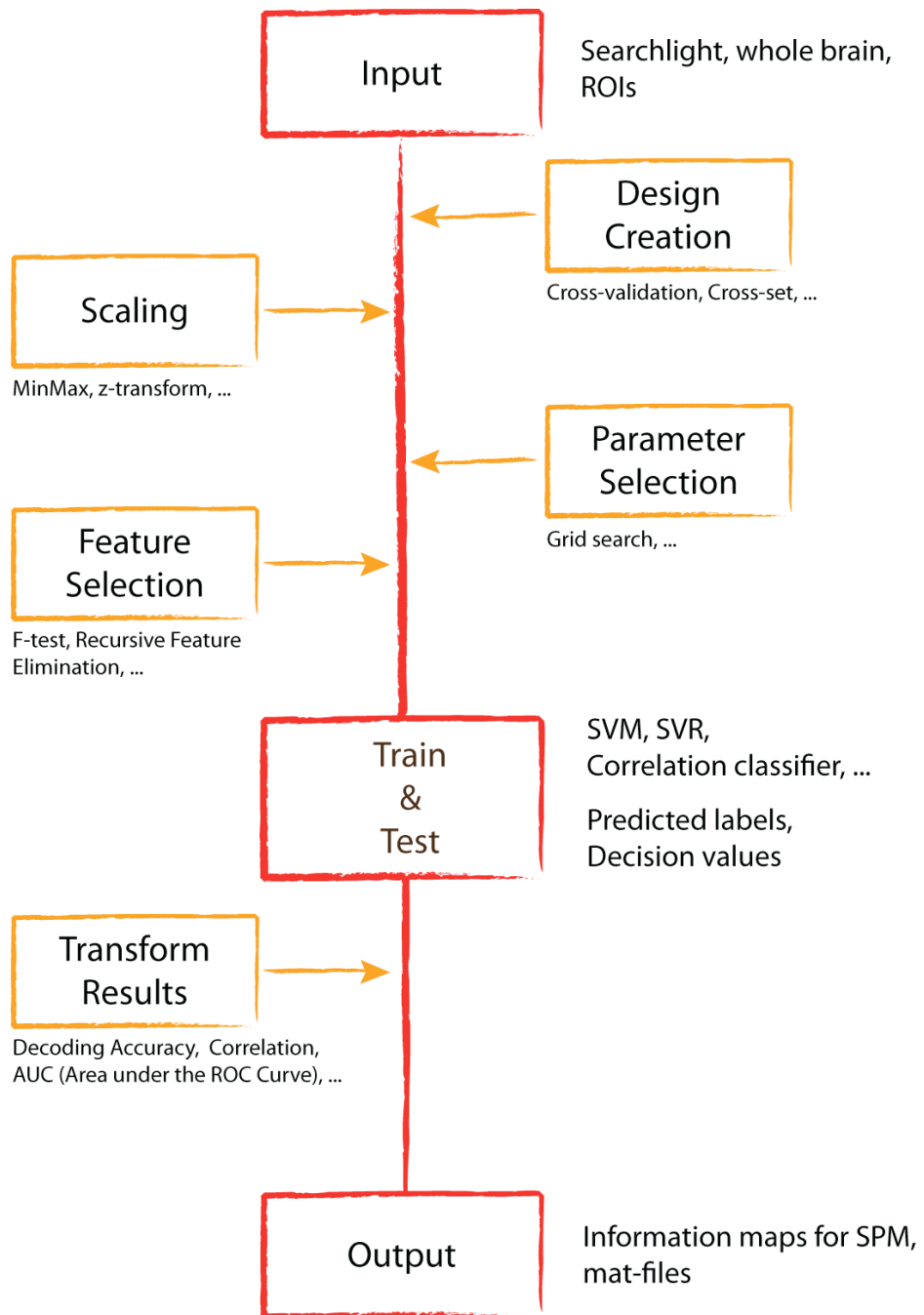
Searchlight-Decoding



Haynes et al., Curr Biol 2007

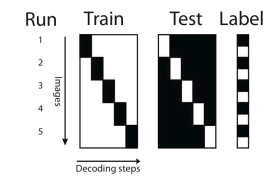
A Standard Decoding Analysis



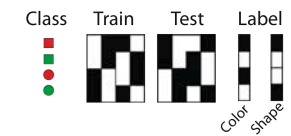


Designs

Standard CV



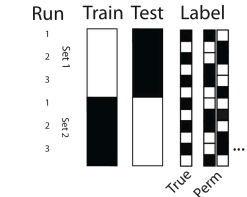
Cross set multi-feature



Cross set



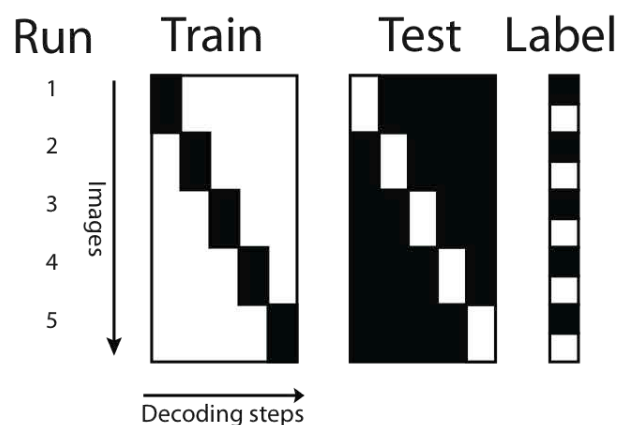
Permutation



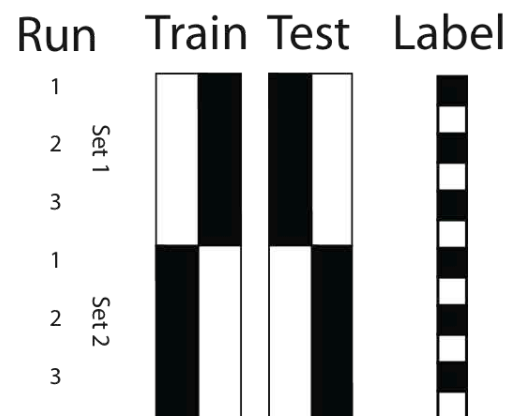
Decoding Accuracy, Correlation, AUC, ...

Different Decoding Designs

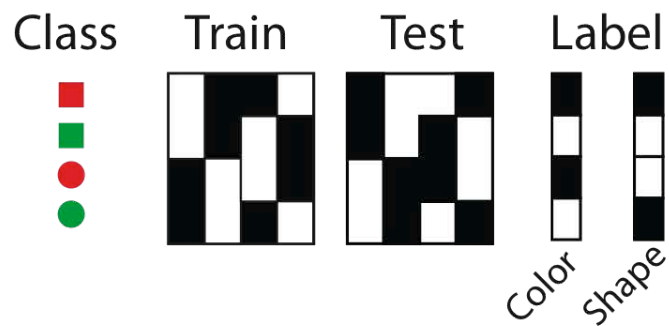
Standard CV



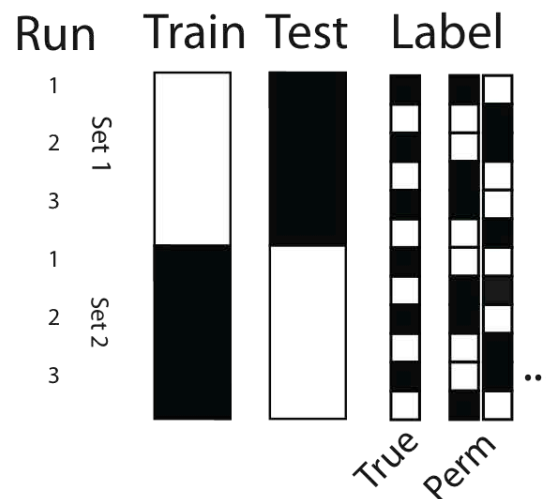
Cross set



Cross set multi-feature



Permutation



Some remarks

- Apply **Multiple-Comparison-Correction!**
 - e.g. in SPM: use FWE-correction on cluster-level
- Over-fitting **during training** make your analysis not wrong if you use an **independent test set**
 - e.g. Cross-validation, Cross-set-decoding
- Circular Analysis
 - The ‘Voodoo’ problem
 - No problem for Decoding with ROIs or Searchlight
 - An example for how a circular analysis could occur:
Perform a Searchlight analysis first, the extract ROIs from this and apply them at the same positions
- Social brain responses in a dead fish
 - No problem, if you don’t correct

TUTORIALS

Useful Software

For the tutorial

- The Toolbox
- MRICron & MRICroGL: Basic image viewer
- testdata

In general

- SPM: General fMRI & M/EEG analysis framework

Toolbox 'Installation'

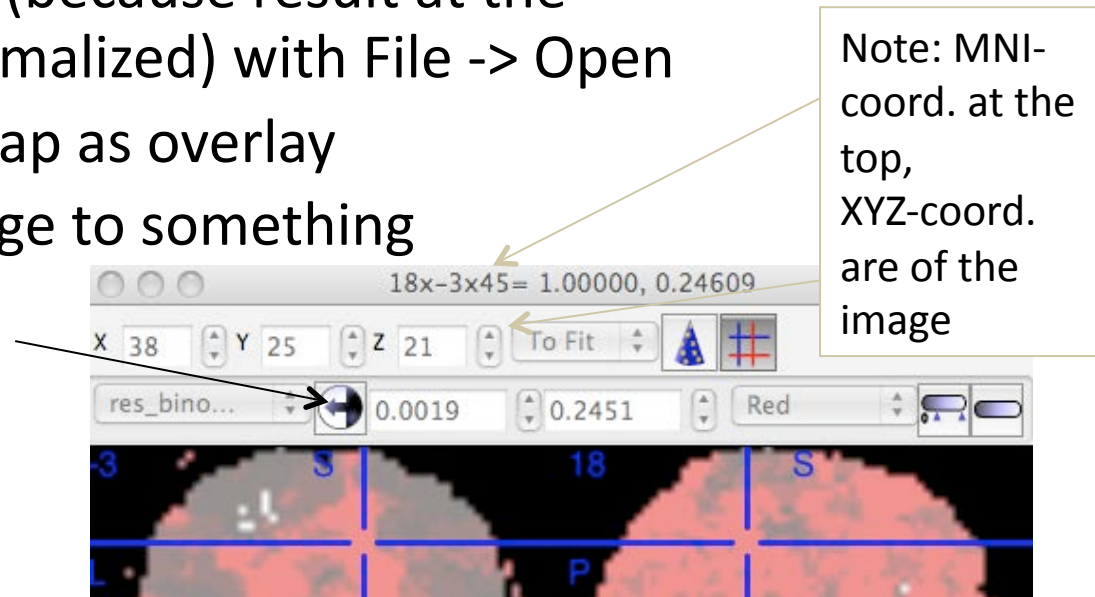
- Extract the .zip archive to some folder
- Add this folder to your matlab path
 - `addpath('c:\decoding_toolbox');`
- Check that it works calling
 - `cfg = decoding_defaults`

Getting started: decoding_tutorial.m

- Open decoding_tutorial.m
- Fill in all required fields
 - labels: 'left' & 'right'
 - left: left button, right hand index finger
 - right: right button, right hand index finger
- Run
 - should take about 3 mins

Looking at your results with MRICron & MRICroGL

- General
 - Load brain mask (because result at the moment not normalized) with File -> Open
 - Load accuracy map as overlay
 - **CHANGE** the range to something meaningful
 - e.g. 60 – 100 for accuracies



- MRICron shows values of overlays in the header (behind MNI coordinates)
- MRICroGL has a nice real-time rendering

General Explanation

- `cfg`
 - Important structure that contains all required information for the decoding
- `decoding(cfg)`
 - Function that does the work
 - Lot's of explanations about `cfg` settings at the top
- `cfg = decoding_defaults()`
 - All default settings + some instructions

Example 2 – Toy Data Decoding

- Open `demo\IK_demo2_toydata.m`
- Run
- Tasks
 - Plot the hyperplane and the margins (i.e. get the model parameters as output measure)
 - Change SVM decoding parameter costs back to default (1) and see what happens to the boundary
 - Get the primal weights and plot them

Example 3: How to add a result transformation

- Have a look at HOWTOEXTEND.txt in the decoding base folder and write a short result transformation that simply checks if the last prediction was correct

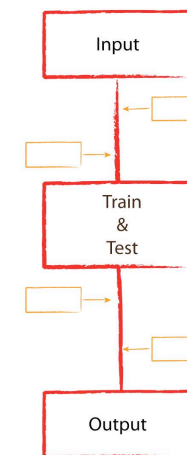
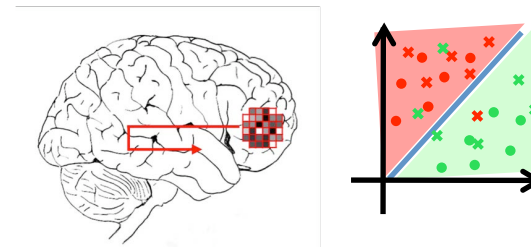
The Decoding Toolbox

How to get & Alternatives

- How to get The Decoding Toolbox?
 - Create a sourceforge account and e-mail us your login
- We provide on Sourceforge
 - Regular updates, Wiki, Room for questions
- If you have questions
 - e-mail us
- Current status
 - About 20 active users
 - Everything (seems) to work fine so far
 - Thus we are optimistic that it will do so in the future, too
- Alternatives
 - pyMVPA (Python, active community)
 - Princeton Decoding Toolbox (matlab, development stopped)

Take Home

- fMRI detects new cars
- Decoding + Searchlight 'increases' the sensitivity of data analysis by looking for characteristic patterns
- The Decoding Toolbox makes decoding easy



Thanks to

Martin Hebart
John-Dylan Haynes

Thank you for your attention

Questions?

Contact: Kai.goergen@bccn-berlin.de

**FURTHER THINGS TO PLAY WITH
(IN CASE YOU LIKE)**

Things to play with

- Visualize the outcome of different 2d SVMs
 - RBF, polynomial, ...
- Get the histogram for different output measures for random / separated data
 - re-run data simulation & analysis a couple of times & draw the outcome distribution
 - measures: decoding_accuracy, AUC, ...
- Perform a multi-cross-set decoding analysis
 - simulate data
 - create design matrices (train, test, label)
 - run it & analyse it's result
- Implement a permutation test